

ChE 382: Unit Operations Laboratory

CONTINUOUS DISTILLATION

Spring 2024v1

Introduction

Distillation is the most widely used separation process in chemical engineering. It can handle a wide variety of feeds and has the potential to produce very high purity products. At times, a petrochemical plant appears to consist mainly of distillation columns involved in product purification. When a chemical engineer faces a separation problem in process design, the usual first alternative is distillation.

Distillation is classified as an equilibrium separation process since the gas and liquid phases are expected to reach phase equilibrium on each contacting stage of the distillation column. (The other major class of separations is rate-governed processes such as filtration, which depend upon the relative rates of flow of the components under a driving force).

Equilibrium on each stage is a theoretical concept which cannot be totally achieved in practice. Chemical engineers have developed models for distillation which consider this non-ideal behavior. Thus, the concept of efficiency is introduced into the analysis of the separation process.

In this experiment, you will examine a **Binary (two component) Distillation Column** that is separating ethanol and water by operating at steady state with a continuous feed, a top product reflux ratio and a bottom product. A good review of distillation technology and theory can be found in the references [1] – [5].

Basic Theory

The basic process concept for a continuous distillation column is shown below. The feed enters near the middle of the column. Heat is supplied in the reboiler to provide the energy to boil the mixtures. The product vapors are condensed at the top of the column. Within the column are the stages where gas and liquid contact; the liquid flowing down the column, the vapor up the column. Ideally, there is vapor-liquid equilibrium on each stage of the column

An important process variable is the reflux ratio at the top of the column. This plays an essential role in controlling the product purity and in determining the energy which is required in the reboiler. Also of importance is the condition of the feed stream (e.g., saturated liquid) entering the column.

A complete simulation of the distillation column would include the following balances:

1. Overall mass balance for the column. A mass balance for each component and the overall balance.
2. Mass balance for each of the equilibrium stages.
3. An overall heat balance for the column.
4. A heat balance for each stage. This may need to take into account the heat of mixing of the components.

This can become a very complex set of equations for a multi-component distillation column. There are numerous sophisticated computer programs available to analyze these columns to let the chemical engineer devise an operational and efficient design. Examples of design packages widely used are CHEMCad, HYSYS, and Aspen. Many companies have their own proprietary design tools.

McCabe-Thiele Analysis for Binary Distillation

For a binary system, the classic McCabe-Thiele graphical technique can be used to design the distillation column and to explore the effect of process variables. Although it is quite simple to use, this simplicity comes with a price. The technique uses a number of assumptions concerning the operation of the distillation column. It is important to analyze whether these assumptions are reasonable for the batch distillation unit being used.

To use the McCabe-Thiele analysis, one needs the vapor-liquid equilibrium for the system plotted as a y versus x diagram. The second part is the operating line for the column which is a formulation of the mass balance from stage to stage; this requires the overall gas and liquid flows and the selected reflux ratio. When the distillation is continuous and feed enters the middle of the column, there are separate operating lines (mass balances) for the column above and below the point of feed introduction. For batch distillation, there is only one operating line since the molar flows of gas and liquid are constant throughout the column.

One may examine the assumptions involved in deriving this operation line. The number of stages is determined by moving from equilibrium to operating curve such as occurs in the column. This generates a step structure which gives the number of stages. While the method is simple to do in practice, it takes some analysis to understand the concepts.

A typical McCabe-Thiele diagram is shown below:

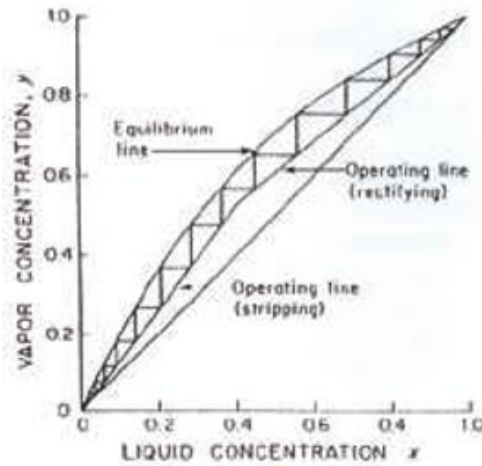


Figure 2. McCabe-Thiele diagram.

Column Efficiency

The basic McCabe-Thiele method assumes vapor-liquid equilibrium on each stage. Many chemical engineers have invested major efforts in devising mechanical devices to promote the attainment of equilibrium on the stage. However, this equilibrium is never obtained completely in practice and in many situations the assumption of equilibrium is a poor one.

The efficiency is introduced with the concept of Murphee plate efficiency, which is defined as:

$$\eta = \frac{y_n - y_{n+1}}{y_n^* - y_{n+1}} \quad (1)$$

Where y_n^* is the composition in equilibrium with X_n . If the vapor and liquid are in thermodynamic equilibrium, Y_n equals Y_n^* . The figure below shows the practical effect on the McCabe-Thiele diagram; inclusion of the plate efficiencies lowers the equilibrium curve to the dashed line.

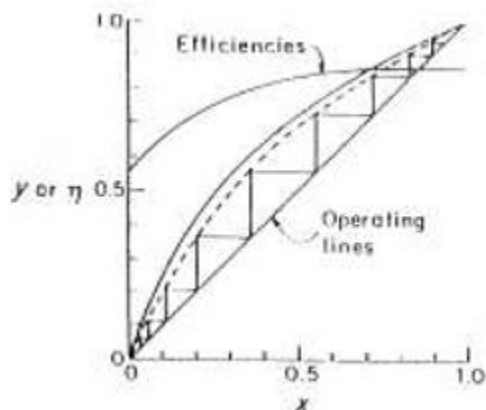


Figure 3. McCabe-Thiele analysis

Armfield Distillation System

The Armfield UOP3CC Continuous Distillation Column is a self-contained distillation facility consisting of two interconnected units; a floor-standing process unit and a bench-mounted control console. A PC computer is also connected to the console. The column can be operated from the console or the computer using Armfield software. **NOTE: Before running the experiments, read over the Armfield Instructional Manual [6] to become familiar with the equipment, operation procedures and safety guidelines.**

Experimental Procedure: Overview

Two ethanol-water solutions will be used for the experiments:

Reboiler solution: 10 liters, 10 wt% ethanol, 90 wt% water

Feed solution: 3 liters, 20 wt% ethanol solution, 80 wt% water

Two experiments will be performed:

Experiment E: Steady State Distillation of a Binary Mixture

Experiment F: Varying the Feed Position: Steady State Distillation of a Binary Mixture.

200-proof ethanol is available in 1-gallon bottles the fume hood on the south lab wall in EIB 128. Read the MSDS for ethanol to understand the hazards and handling procedures. Fill the reboiler with the 10 wt% ethanol solution to 10 liters and one of the feed tanks with 3L of the 20 wt% ethanol solution. Do not turn on the distillation column until you check with TA or instructor. You can monitor the progress of the distillation column by watching the tray temperatures as a function of time using the control console or the software. Each tray will warm up as liquid accumulates on the tray. These temperatures can be used to estimate the composition of the liquid on the tray assuming that the trays are at equilibrium. Use these temperatures to examine the operation of the column, and to monitor its approach to steady-state operation. Vapor-liquid equilibrium for ethanol-water is available in literature.

Experiment A: Feed Pump Calibration

The peristaltic feed pump is designed to give approximately 1 ml/min per revolution of the drive shaft. As the variable speed motor is capable of speeds varying from 0 to 300 RPM, the pump will be able to deliver approximately 0 to 300 ml/min of feed to the column.

In order to achieve greater accuracy, it is necessary to produce a calibration graph of the actual flowrate against the position of the variable speed dial on the control console. The dial has ten full turns, each full turn marked in one hundredth segments. To produce the graph, it will only be necessary to measure the flow at ten settings over the full range.

Disconnect the feed tubing to the distillation column. Ensure the feed tank with the pump suction pipe inserted has sufficient water for the calibration. Using a graduated cylinder and a stopwatch, determine the flowrate at the ten settings and construct a graph which can be subsequently located near the control console.

Experiment B: Steady State Distillation of an Ethanol-Water Mixture

Objective

To investigate the steady state distillation of a binary mixture under continuous operation.

Method

The column will operate with a reflux ratio of 5:1, a feed position in the middle of the column at steady state, and a bottom product stream. Top product and bottom product liquid samples will be taken to determine the ethanol composition. A McCabe-Thiele analysis will be performed, and the overall column efficiency will be calculated.

Equipment Set Up

Make up 10 liters of a mixture of 10 wt% ethanol/water and 3 liters of a mixture of 20 wt% ethanol/water. Set the equipment to operate at total reflux by switching off the reflux ratio timer on the console. Refer to Armfield Instruction Manual for a diagram of the process unit and location of the unit components (valves, etc.). Ensure all valves on the equipment are closed. Open valve V10 on the reflux pipe.

Load the 10 liter of ethanol/water mixture through the filler cap on top of the reboiler using a funnel. Make sure the filler cap on the top of the reboiler is firmly replaced after filling.

Fill one of the feed bottles with 3 liters of the 10 wt% ethanol/water solution. Connect the blue cap to the bottle and connect the black tubing from the feed bottle to the feed pump per the instructions below:

Feed Pump

The peristaltic feed pump (7) has a pivoting clamp on the pumphead for fitting the flexible tubing. To install the flexible tubing, pivot the clamp (on the top of the pumphead) upwards and backwards to expose the rotor. Load the tubing into the slot between the rotor and pumphead then close the pumphead by pivoting the clamp forwards and downwards. The clamp will click shut. The tube will retain its elasticity for many hours of use but there is sufficient length to allow the tube to be moved to other positions of less wear. It is essential to release the clamp when the equipment is not being used to prevent permanent deformation of the tube.

Connect the clear feed tubing from the pump outlet to the column between trays 4 and 5.

Procedure

Turn on the power to the control panel.

Set the temperature selector switch to T9, the temperature in the reboiler, and open valve V5 admitting the cooling water to the condenser at a flow rate on FI1 of approximately 3 liters/min. This rate may vary according to the temperature of the water. If there is insufficient water flow, some of the vapor will be uncondensed and emerge from the vent pipe at the side of the unit in which case increase the water flow rate.

On the control panel turn the power controller for the reboiler heating element fully anti-clockwise and switch the switch turning on the power to the heating element to “power on” position. Another red lamp will illuminate indicating the heating element is on. Turn the power controller clockwise until a reading of approximately **1.5 kW** is obtained on the digital wattmeter. The contents of the reboiler will begin to warm up and this can be observed on the temperature readout meter.

Eventually, vapor will begin to rise up the column and the progress of this can be clearly observed as well as detected by the increasing temperatures when switching the temperature selector on T8, T7, T6, T5, T4, T3, T2 and T1. Vapor will enter the condenser and reappear as droplets into the glass walled distillate receiver vessel. The distillate will build up a small level in the receiver and eventually overflow to the reflux regulator valve. Turn the reboiler power controller to 0.75kW and start the experiment with total reflux, meaning the condensed vapor will return to the column.

When the column has stabilized at total reflux (it takes 15 to 30 minutes), the bottom product flow stream, the flow of feed, and the reflux can be started.

Before switching on the reflux switch, set the reflux ratio to 5:1, meaning 5 sec back to column and 1 sec to top product receiver, (see Operational section). C- means back to column and C+ means to top product receiver. When this is done, switch the reflux valve on (on the control panel). You should now hear a click, meaning the reflux valve is working. You should also see condensed vapor flowing to the top product receiver.

Open valve V1 to let the bottom product exit through the heat exchanger and into the bottom product receiver. The feed pump can now be turned on and the feed to the column will now be admitted to the midpoint of the column. It is advisable to set a feed flow of 30-35 milliliters/min (from the feed pump calibration graph).

After feeding approximately 400 milliliters, take a sample of the overheads through valve V3. In a similar manner **and preferably at the same time** take a sample of the bottom product through valve V2. Take two more samples of the top and bottom product, 5 minutes apart. Also, record the temperatures in the column; T1-T9.

Record the concentration for the samples using the densitometer (SNAP 41). **CAUTION! THE SAMPLES WILL BE HOT.**

After the sampling has been completed, turn off the feed pump, turn off the reflux switch, turn off the reboiler power, and close valve V1.

Experiment C: Varying the Feed Position

Objective

To investigate the effect of varying the feed position under continuous operation.

Theory

The column is provided with three feed positions, one above the top plate, one between the two four-plate sections (mid-point) and one below the bottom plate. In this way the column can be run as a conventional distillation column (feed between the sections), as a rectifying column (feed below the bottom plate) or as a stripping column (feed above the top plate). As the rectifying case is closely approximated by the batch distillation done in the Fall semester, the stripping case (feed above the top plate) is chosen for this experiment.

Changing Feed Position

Disconnect the feed tube from the column at the midpoint and reconnect the plug. Be careful as the column and tube will be hot and some liquid may leak out. You may want to turn off the reboiler power temporarily at this point which changing the feed position.

Disconnect the plug from the feed position above the top plate and connect the feed tube to the column at this position.

Procedure

Turn on the reboiler power to 0.75kW. When the column has stabilized at total reflux, about 15-30 minutes, the bottom product flow, the flow of feed and the reflux can be started at the same time. Since the column is hot, total reflux stabilization should not take as long as in Experiment B.

Before switching on the reflux switch, set the reflux ratio to 5:1, meaning 5 sec back to column and 1 sec to top product receiver, (see Operational section). C- means back to column and C+ means to top product receiver. When this is done, switch the reflux valve on (on the control panel). You should now hear a click, meaning the reflux valve is working. You should also see condensed vapor flowing to the top product receiver.

Open valve V1 to let the bottom product exit through the heat exchanger and into the bottom product receiver. The feed pump can now be turned on and the feed to the column will now be admitted to the midpoint of the column. It is advisable to set a feed flow of 30-35 milliliters/min (from the feed pump calibration graph).

After feeding approximately 400 milliliters, take a sample of the overheads through valve V3. In a similar manner **and preferably at the same time** take a sample of the bottom product through valve V2. Take two more samples of the top and bottom product, 5 minutes apart. Also, record the temperatures in the column; T1-T9.

Record the concentration for the samples using the SNAP 41 densitometer. **CAUTION! THE SAMPLES WILL BE HOT.**

After the sampling has been completed, turn off the feed pump, turn off the reflux switch, turn off the reboiler power, and close valve V1.

Data Analysis

Unsteady State Analysis

Plot the temperature recorded as a function of time.

Briefly explain these temperatures. How does the column approach steady state?

Steady State Analysis

Perform an ideal McCabe-Thiele analysis using equilibrium data for the two feed positions. Use the product and feed compositions you measured, and the reflux ratio you used.

Assume that the feed is a saturated liquid. Compare the results of this analysis with your experimental results. Determine the column efficiency for the two feed position experiments.

Explain the results of your McCabe-Thiele analysis. In particular, how is the McCabe-Thiele analysis inappropriate for this particular distillation column? Which assumptions are valid, and which are invalid? Can you explain your results by assuming that the trays are not 100% efficient? Can you explain your results by assuming that energy is lost in the column? To do this, perform an overall energy balance on the column using the reboiler and condenser heats. Assume that the energy carried away in the two product streams balances the energy added in the feed stream.

References

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- [3] W. McCabe, J. Smith, and P. Harriot, *Unit Operations of Chemical Engineering*, 7th ed. McGraw-Hill, 2005.
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- [5] P. C. Wankat, *Equilibrium-Staged Separations*. Prentice Hall, 1988.